

- ◆ Using the above command sequence for rotation of motor, a program must be made to steer the vehicle.
- ◆ The RF module (transmitter and receiver) will need to be attached to the Arduino™ board to send and receive instructions to speed up, steer and stop the toy car.
- ◆ To instruct the Arduino™ to perform an operation, you'll send single characters over the RF that represent the operation to be performed (such as the common gaming keys mentioned in the table). You can use the 'switch ... case' or an 'if ... else' type of programming to detect the command sent by the user over RF to perform the steering.
- ◆ Assemble the chassis, fix the motors and tyres, and you're done.

Such vehicles can be used for surveillance or during emergencies to assess safely of places without causing damage to human life.

You could also control the car from a smartphone by sending instructions via Wi-Fi or Bluetooth. To achieve this, you'll need an app on your phone that can connect to the car over one of these wireless networks. The app will send the steering information to the car. This can be thought of as real world gaming with toy cars using a smartphone in place of a controller.

Autonomous wheeled robot

This autonomous wheeled robot is a modification of the wireless computer-controlled toy car. It senses a wall and surrounding obstacles to navigate around a room. This can be achieved by using ultrasonic sensors that can check the distance of the robot from the object in front of it.

To make an autonomous robot, you'll need the following components:

- ◆ Motors
- ◆ Wheels
- ◆ Chassis
- ◆ RF transmitter and receiver
- ◆ Battery
- ◆ Motor shield
- ◆ Ultrasonic sensor

How to implement:

- ◆ Use the Ultrasonic sensor to read the distance directly in front of the robot.
- ◆ The Ultrasonic sensor sends an ultrasonic wave (sound inaudible to humans). This ultrasonic wave reflects from obstacles directly in front of the sensor. The program on the Arduino™ board measures the time of